

Phase 13 — Entity Lifecycle Management: Engineering Report

Status: COMPLETE **Dates:** 2026-07-05 to 2026-07-07 **PRs:** #94, #95, #96, #97, #98, #99, #100, #101, #102, #103, #104 **ADRs:** ADR-0034 (new; revised twice; supersedes ADR-0017's PURGE mechanism for the on-demand deletion case; ADR-0017 itself unmodified, still accepted)

Sub-phase summary

Sub-phase	PR	Scope	Key finding
13.0	#94	Deactivate bug fix	active field never set on user deactivation; frontend filter relied on it
13.1	#95	Audit logging + cancellation notifications + facility deactivate enhancement	cancel_booking sent zero notifications for any cancellation, ever — fixed globally
13.2	#96	Direct permanent delete (residents/tenant-admins)	New DELETE .../permanent route; ConfirmDialog gains reusable type-to-confirm
13.3	#97	Facility delete replaces deactivate; hide Users Deactivate; harden delete	Driven by a live incident: deactivate-without-reactivate created an unrecoverable stuck Firebase Auth account
13.4	#98	Tenant cascade delete, platform-admin only	recursive_delete + independent Auth cleanup + platform_deletion_log top-level collection (survives its own cascade)
13.5	#101	Search, admin visibility, mobile layout, /version, temp-password modal, CSV styling, bulk-import tests	Bulk-import endpoint had zero prior test coverage
13.7	#99	Apex domain support + cross-tenant/apex sign-in redirect	Apex DNS/LB (infra, direct-applied) + unified AuthContext fix for both bugs
13.8	#100	Force token refresh + claims-error recovery UI	Firebase custom claims propagate on next refresh, not immediately

—	#102	Terraform check-in (apex host_rule)	Infra was live since 13.7a but never committed — found during closeout audit
—	#103	Booking Policies fetch-on-mount fix	No GET /tenant/policies route ever existed; form always showed hardcoded defaults
—	#104	Terraform: route /version through LB	/version (13.5) was never added to the LB path_matcher

Metrics

- Backend: 410 tests, 91% coverage (gate: 90%)
- Frontend: 314 tests, 41 files
- New top-level Firestore collection: platform_deletion_log
- New backend routes: DELETE /tenant/users/{uid}/permanent, DELETE /tenant/facilities/{id} (behavior changed, path unchanged), DELETE /admin/tenants/{tenant_id}/permanent, GET /tenant/policies, GET /version
- New frontend capability: ConfirmDialog type-to-confirm (confirmationPhrase prop), TempPasswordModal, AuthContext's first-ever test file

Real incidents encountered and resolved during the phase

1. **Stuck Firebase Auth account.** A resident deactivated before Phase 13 blocked re-registration of their own email indefinitely, with no UI recovery path. Root cause of the 13.3 design reversal (deactivate → delete for facilities; hid Users Deactivate pending a proper Reactivate design).
2. **Apex domain unreachable + silent cross-tenant sign-in.** No DNS/LB routing existed for the bare domain; once added, sign-in from the wrong subdomain (or the apex) succeeded without redirecting to the correct tenant. Both fixed in 13.7.
3. **Stale-token 401 on fresh sign-in.** Custom claims timing caused a silent, permanent blank screen on the very first post-sign-in API call for some accounts. Fixed in 13.8; introduced the codebase's first client-side auth test coverage as a side effect.
4. **CI catch on PR #100.** A required field added to a shared AuthState type broke three other test files' mocks, outside the fixing prompt's declared file scope; caught by CI's tsc -b step after the change's own test suite had already reported 100% pass. Root cause: a test runner does not project-wide type-check the way a build step does. Documented as protocol \$5.30.
5. **Two bookkeeping gaps found during phase closeout, not before.** Terraform drift (infra live, git not updated) and a missing LB route for /version (the endpoint built specifically to diagnose exactly this class of problem). Both closed via #102 and #104.
6. **13.5's facility mobile fix was necessary but not sufficient.** ListRow (shared by Facilities, admin Tenant List, and My Bookings) is unconditionally row-layout with a fixed-width action area; removing truncate alone left names wrapping one word per line under 3 action buttons. Fixed at the shared-component level once live-device testing (after the phase was believed closed) showed the original fix was incomplete.

Deliberately deferred (not incomplete — decided)

- **13.6, CDN cache-fill investigation.** Reproduced once (a cold subdomain path returned a 0-byte body after the apex-redirect flow); root cause unconfirmed after investigation (no Age header on any test, undermining but not disproving the CDN-caching theory). Coordinator decision: not currently recurring, backlogged rather than pursued further without stronger evidence.
- **Deactivate/Reactivate as a paired feature.** Coordinator explicitly chose Delete-only simplicity for the current phase; proper reversible-deactivation UX is future work, not lost scope.
- **Bulk-import reason field inconsistency** (human message vs. machine code) between the admin and tenant-admin bulk-create endpoints — documented in the 13.5 test suite, tracked, not unified.

ADR-0034 revision history (within this phase)

1. Facility lifecycle (adapted 3-stage pattern) + resident DPDP erasure via full delete.
2. Generalized to a direct, independent Delete action for every entity type (residents, tenant-admins, tenants) — Coordinator's stated preference for simplicity over ADR-0017's slow admin-purge model.
3. Facility-specific scope further simplified from "deactivate + future PURGED stage" to "delete only," following the live stuck-account incident (13.3).

ADR-0017 remains accepted, unmodified. ADR-0034 accepted at phase close.

Protocol changes during this phase

- v3.1 → v3.3 (this project's incremental track): §5.23 (pin toolchain versions), §5.24 (never silence auth/permission stderr), §5.30 (shared-interface ripple + test-suite-passing insufficiency).
- v3.4 adopted wholesale (externally authored, brought in mid-phase): §0 self-certification, §2.1a scoping/sequencing discipline, §4.12 proactivity mandate, §7.6 retrospective authorship, §8 token discipline (consolidated asks, text-over-screenshots, confirm reproduction context first).
- v3.5: §5.30 re-added after a renumbering collision with v3.4's own new §5.25-5.29 entries; final state verified via a clean sequential §5.1-§5.30 listing before sign-off.

Carried into the next phases (not this phase's scope, logged for continuity)

- Daily Booking Overview (tenant-admin day view, accessible tooltips)
- Voice input/output for the AI Booking Assistant (STT → translate → existing Phase 9 agent, unchanged → translate back → TTS)